2. They absorb UV light at 260 nm. This property is used to detect and estimate DNA and RNA in biological solutions.

3. They are capable of forming hydrogen bonds with purine nucleotides in nucleic acids.

4. They exhibit keto–enol tautomerism.

ii. Purine bases

Purine is a bicyclic ring formed by fusion of pyrimidine ring with imidazole ring. Purine ring is numbered in the anti-clockwise fashion. Adenine (6-amino purine) and guanine (2-amino-6-oxy purine), the two common purines, are found in both DNA and RNA (Figure 7.2). Other naturally occurring purine derivatives include hypoxanthine, xanthine, and uric acid. Hypoxanthine and xanthine are found only rarely as constituents of RNA, while they are intermediates (compounds formed in the synthetic pathway) in the synthesis of nucleic acids Uric acid is the catabolic end product of nucleic acids.

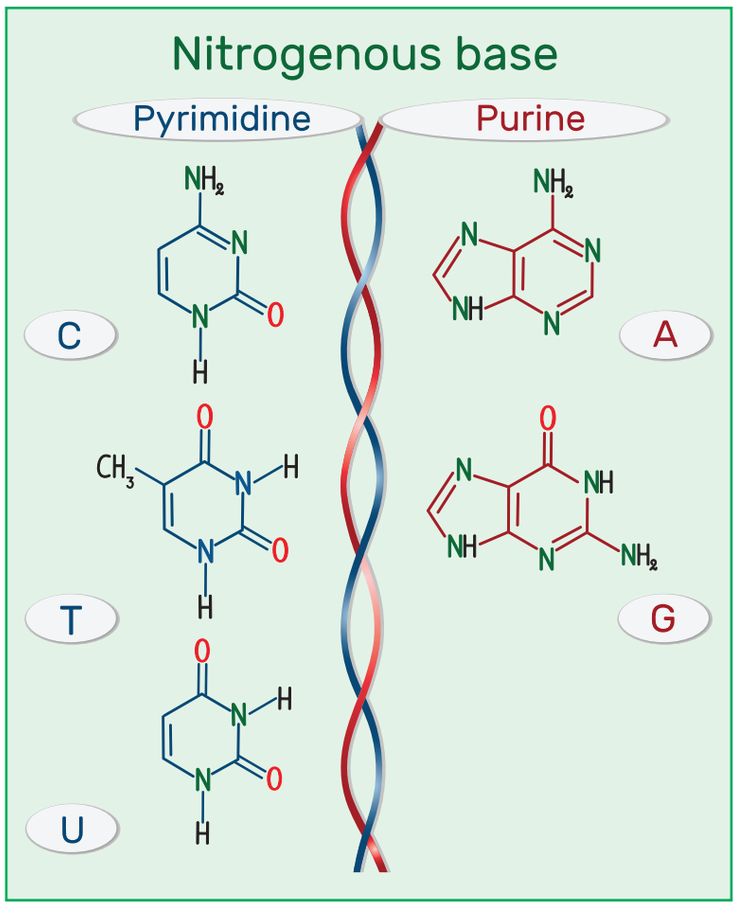


Figure 7.2. Structure of Purine and Pyrimidine bases

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